

# Homework 3: OLS Is a Likelihood

BEE 4850/5850, Spring 2027

Due: Thursday 18 February 2027, 9:00pm Eastern

Submit a PDF to Gradescope with **each sub-part tagged**. Untagged submissions lose 10%.

**20 points.** Problems 1 and 2 are pen-and-paper (14 points). Problem 3 is computational (6 points).

## Problem 1 — The hinge (7 points)

A colleague fits a straight line to the Sewells Point annual maxima from Homework 1, regressing the annual maximum  $y_i$  (m) on years since 1928,  $t_i$ . Their software reports:

|                           | estimate | std. error |
|---------------------------|----------|------------|
| intercept $\hat{\beta}_0$ | 0.9523   | 0.0547     |
| slope $\hat{\beta}_1$     | 0.004363 | 0.000966   |

with  $\hat{\sigma} = 0.2581$  m and  $n = 92$ .

(a) (3 points) Write out the Gaussian-error linear model in full, **including any and all assumptions**. Distinguish clearly between what is assumed about the *errors* and what is assumed about the *predictor*.

(b) (2 points) What does it mean to say that ordinary least squares **maximizes the Gaussian likelihood**? Which assumption is that equivalence relying on? Show the step in the log-likelihood where the assumption enters.

(c) (2 points) Why does adding a distributional assumption to least squares buy you anything at all? **Name two things** you can do with the likelihood version that you cannot do with least squares alone.

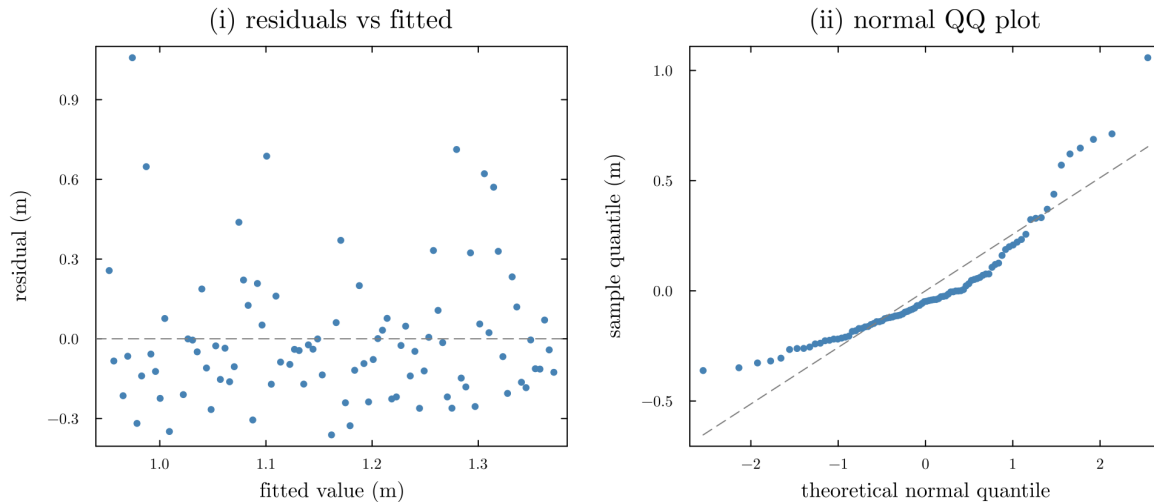
## Problem 2 — Short questions (7 points)

(a) (3 points) Suppose the errors were **Laplace** rather than Gaussian, with density

$$f(e) = \frac{1}{2b} \exp\left(-\frac{|e|}{b}\right).$$

What would the maximum-likelihood estimate of  $(\beta_0, \beta_1)$  minimize instead of the sum of squares? **Derive it from the likelihood** — do not simply assert the answer. In one sentence, say what this implies about the influence of a single large outlier.

(b) (2 points) The two panels below are diagnostics for the fit in Problem 1.



- Which assumption does **each** panel test?
- Which panel shows a violation, and **what specifically in the plot** tells you?

(c) (2 points) The 95% confidence interval for  $\beta_1$  is (0.00244, 0.00628) m/yr. Mark each statement **true** or **false**; correct the false ones in a phrase.

- There is a 95% probability that  $\beta_1$  lies in this interval.
- If the experiment were repeated many times, about 95% of such intervals would contain  $\beta_1$ .
- The interval excludes zero, so the trend is real.
- A wider interval would mean a larger estimated trend.

## Problem 3 — Recovery in miniature (6 points)

Use `data/annual_maxima.csv` (the output of Homework 1), keeping only years with at least 8,000 recorded hours.

(a) (2 points) Fit the regression of annual maximum on years since 1928. Report  $\hat{\beta}_1$  and its standard error to sensible precision.

(b) (3 points) Now **simulate 200 datasets from the fitted model** at the observed  $t_i$  — that is, draw  $y_i^* = \hat{\beta}_0 + \hat{\beta}_1 t_i + \varepsilon_i$  with  $\varepsilon_i \sim \text{Normal}(0, \hat{\sigma}^2)$  — and refit each one. Report the standard deviation of  $\hat{\beta}_1$  across the 200 refits.

**Show the comparison as a figure**, not a table of 200 numbers.

(c) (1 point) In one sentence: *should* (a) and (b) agree, and *do* they?